Hong-Vinh Vu

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About

Control and Automation Engineering undergraduate at HUST with research experience in intelligent control and dynamic optimization (Learning-based control). Aiming to specialize in optimal control and multi-agent reinforcement learning for autonomous robotic systems.

Education

Hanoi University of Science and Technology (HUST), B.Sc. of Automation Engineering. Sep 2024 – Present

- **GPA:** 3.45 / 4.0
- **Relevant coursework:** Modern Control Theory, Deep Learning, Multi-agent System

Le Quy Don High School for the Gifted, Specialized Physics Program. Aug 2021 – May 2024

- Won the Second Prize, Provincial Physics Grade 12 Olympiad.

Academic Experiences

Research Assistant, Mar 2026 – Present
Multi-Agent System Control (MASC) Laboratory, SEEE, HUST.

- **Supervisor**: Assoc.Prof. Hoai Nam Nguyen
- **Research topic**: Applied Optimal Control Algorithms and Intelligent Control Algorithms to optimize trajectory tracking for Quadrotors.
- **Key Contribution**: Developed a hybrid flight control framework by integrating PyTorch-based Deep Reinforcement Learning (DRL) for high-level dynamic trajectory planning with Model Predictive Control (MPC) for optimal trajectory tracking and nonlinear disturbance compensation.

Research Assistant, Mechatronics Engineering Group (MEG), Jul 2025 – Mar 2026
Motion Control and Applied Robotics Laboratory (MoCAR LAB), SEEE, HUST.

- **Supervisor**: Assoc.Prof. Danh Huy Nguyen
- **Research topic**: Kinematic Modeling and Control of a 6WD-6WS Mobile Robot based on ROS2.
- **Key Contribution**: Developed mathematical kinematic/dynamic models and motion control algorithms for a 6WD-6WS (6-Wheel Drive, 6-Wheel Steering) omnidirectional vehicle, implementing custom ROS2 navigation nodes, Gazebo physics simulations, and embedded MCU firmware for real-time trajectory tracking and multi-mode steering coordination.

Personal Projects

Anti disturbance formation control of quadrotor UAVs based on terminal sliding mode control:

Robust tracking control strategy for a quadrotor using RPD-SMC and RISE:

Backstepping Approach for Controlling a Quadrotor Using Lagrange Form Dynamics:

ISSA-Optimized BP Neural Network PID Attitude Controller for Quadrotor: Designed an adaptive BP neural-network PID controller for quadrotor attitude control, using the Improved Sparrow Search Algorithm (ISSA) to optimize the network's initial weights and biases offline for real-time disturbance compensation, improving trajectory-tracking robustness against unmodeled dynamics and parameter uncertainty.

Nonlinear MPC Controller for Quadrotor: Formulated a cascaded NMPC scheme on full 12-state nonlinear dynamics, decoupling attitude and position loops with torques as MVs and thrust as a measured disturbance. Solves a constrained finite-horizon optimal control problem online, predicting trajectory deviation over a receding horizon for anticipatory disturbance rejection.

Fuzzy-PD Controller for Quadrotor: Developed a Self-Tuning Fuzzy-PD controller that dynamically adjusts K_p and K_d gains based on real-time tracking errors, significantly improving dynamic response and overshoot reduction.

Direct Adaptive Fuzzy Controller for Quadrotor: Implemented a Direct Adaptive Fuzzy Control (DAFC) scheme using Lyapunov stability theory to approximate unmodeled dynamics and external disturbances, achieving robust trajectory tracking with minimal steady-state error.

PID Controller for Quadrotor: Designed and tuned a discrete PID controller in MATLAB/Simulink for 6-DOF Quadrotor position and attitude stabilization.

HONORS & AWARDS

Academic Achievement Scholarship, Grade: Excellent. Apr 2025
This is a prestigious award given to students with the highest scores in the entire department ($\approx 5\%$).

TECHNICAL SKILLS

Programming: Python, C, C++, MATLAB/Simulink.
DL / Optimization: PyTorch, Deep Learning.
Robotics Developments: ROS2 framework, Gazebo, SLAM.
Tools: LaTeX, STM32CubeIDE.